

```
--fn_passorfail
select dbo.fn_passorfail(10, 3) as [student 10 exam 3 result];
go
--fn_countcorrectanswers
select dbo.fn_countcorrectanswers(10, 3) as [should be >= 0];
go
```

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Results		Messages	
student 10 exam 3 result:			
1	PASSED		
should be >= 0			
1	3		

```
--fn_isusernameavailable
select dbo.fn_isusernameavailable('admin01') as [should be 0 - taken],
       dbo.fn_isusernameavailable('totallynewuser999') as [should be 1 - available];
go
```

	should be 0 - taken	should be 1 - available
1	0	1

```
--fn_studentexamhistory (tvf)
select * from dbo.fn_studentexamhistory(10) order by ex_date desc;
go
```

	ex_id	crs_name	ex_type	ex_date	full_mark	student_mark	grade	result	actual_start_time	actual_end_time
1	3	OOP	exam	2025-10-01	100	50.00	D	PASSED	2025-10-01 10:00:00.000	2025-10-01 12:00:00.000

```
--fn_examscoreboard (tvf)
select * from dbo.fn_examscoreboard(3) order by rank_position;
```

	rank_position	st_id	st_name	student_mark	full_mark	percentage	grade	result
1	1	10	Leen	50.00	50.00	100.00	D	PASSED
2	2	9	Nedal	20.00	20.00	100.00	F	FAILED

\

TEST Queries

```
select * from vw_instructor_courses;  
go
```

	ins_name	crs_name	in_id
1	Ahmed Mohammed	OOP	1
2	Omar Khaled	C#	2
3	Nour Hassan	LINQ	6
4	test instructor	OOP	11
5	test instructor	test course	11

```
select * from vw_students_per_intake;  
go
```

	in_name	total_students
1	Intake 41	1
2	Intake 42	1
3	Intake 43	1
4	Intake 44	1
5	Intake 45	1
6	Intake 46	1
7	Intake 47	1
8	Intake 48	1
9	Intake 49	1

```
select * from vw_top_students;  
go
```

	ex_id	st_name	total_score
1	3	Leen	50.00
2	4	updated name	25.00

```
--get exam details
declare @exid int = (select top 1 ex_id from exams
                    where ins_id = (select ins_id from instructors where
                                   order by ex_id desc);
exec sp_get_exam_details @exid;
go
```

	ex_id	ex_type	crs_id	ins_id	in_id	ex_date	start_time	end_time	total_time	total_degree	year	allowance_options	crs_name	ins_name	in_name
1	4	exam	11	16	11	2025-12-01	09:00:00.0000000	11:00:00.0000000	120	100	2025	NULL	test course	test instructor	test intake 99

```
--search exams
declare @cid int = (select crs_id from courses where crs_name = 'test course');
exec sp_search_exams @crs_id = @cid;
go
```

	q_id	q_text	q_type	q_degree
1	20	what is object?	mcq	30

```
--get exams by course
declare @cid int = (select crs_id from courses where crs_name = 'test course');
```

	ex_id	ex_type	ex_date	year	total_degree	total_time	crs_name	ins_name	in_name
1	4	exam	2025-12-01	2025	100	120	test course	test instructor	test intake 99

```
283 -- add question with invalid type -> should fail
284 declare @cid int = (select crs_id from courses where crs_name = 'test course');
285 declare @insid int = (select ins_id from instructors where ins_email = 'testinstructor@test.
286 begin try
287     exec sp_add_question 'bad question', 'essay', 'answer', @cid, @insid;
288     print 'error: should have failed!';
289 end try
290 begin catch
291     print 'passed: ' + error_message();
292 end catch;
293 go
294
```

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Messages

passed: invalid question type. use mcq, truefalse, or text.

Completion time: 2026-03-31T22:51:24.6366016+02:00

```
select * from vw_failed_students;  
go
```

	st_name	crs_name	total_score	min_degree
1	Nedal	OOP	20.00	50
2	updated name	test course	25.00	50

```
select * from vw_course_success_rate;  
go
```

	crs_name	success_rate
1	OOP	50.00000000000000
2	test course	0.00000000000000

```
select * from vw_exam_overview;  
go
```

	ex_id	crs_name	ins_name	ex_date	total_degree	total_time
1	3	OOP	Abdelrahman Said	2025-10-01	100	120
2	4	test course	test instructor	2025-12-01	100	120

```
Permissions_Roles...SRAA ♥ \Hany (86)) Test.sql - (local)...(ESRAA ♥ \Hany (68))* Esraa ♥ .Examnatio...training_managers
630
631 -- module 10: triggers
632 --tr_checkexamtime - answer outside exam time window
633 begin try
634     insert into student_answers (st_id, ex_id, q_id, student_answer, is_cor
635     values (10, 3, 5, 'unique identifier', 0, 0);
636     print 'note: trigger allows (exam window check based on getdate vs exam
637 end try
638 begin catch
639     print 'passed: ' + error_message();
640 end catch;
641 go
```

149 %

Messages

(0 rows affected)
passed: Violation of UNIQUE KEY constraint 'UQ__student__360EE635D91651CE'. Cannot insert duplicate key in c
Completion time: 2026-03-31T22:45:04.0785420+02:00

149 %

Overwritten successfully: ...